Vasishth Intro to Frequentist Statistics

Foundations

What is a *Random Variable*?

- sample space: different possible outcomes associated with a probability
- a random variable X, a function that associates to each outcome $\omega \in S$ exactly one number X(ω) = x (for example, number of tosses until one gets a head (H))
- S_x is all the x's (all the possible values of X, the support of X)

Example: coin tosses till H

- $X : \omega -> x$
- ω : H, TH, TTH,... (infinite)
- $x = 0, 1, 2, ...; x \in S_x$

And $X(\omega) = x$: $H -> 1$, $TH -> 2$,...

Probability Distribution Functions

- every discrete random variable X has associated with it a *probability mass function (PMF)*
- continuous RVs: *probability density functions (PDFs)*
- $px : S_x$, $x S_x

Cumulative distribution function: cumulative probability of getting a heads in 1 or less, 2 or less tosses,...

Binomial random variable

- toss coin 10 times, two possible outcomes, number of successes
- FORMULA

Calculate the probability of P(X <= x), seeing n heads, using cumulative distribution function

```
n <- 10 # sample size
p <- 0.5 # probability of success

probs <- rep(NA, 11)
for (x in 0:10){
    probs[x+1] <- round(pbinom(x, size=n, prob=p), digits=2)
}

data.frame(p = seq(0,10), p = probs)
```

```
    p  p.1
1   0 0.00
2   1 0.01
3   2 0.05
4   3 0.17
5   4 0.38
6   5 0.62
7   6 0.83
8   7 0.95
9   8 0.99
10  9 1.00
11 10 1.00
```

10 has to be one because it encompasses all possible outcomes (n=10). the distribution is cumulative: if 2 heads, you can add probability of getting 0,1 and 2 heads,...

discrete data!

Probability of getting exactly 1 head:

```
choose(10,1) * 0.5 * (1-0.5)^9
```

```
[1] 0.009765625
```

```r
dbinom(1, size=10, prob=0.5)
```

```
[1] 0.009765625
```

Generate random binomial data

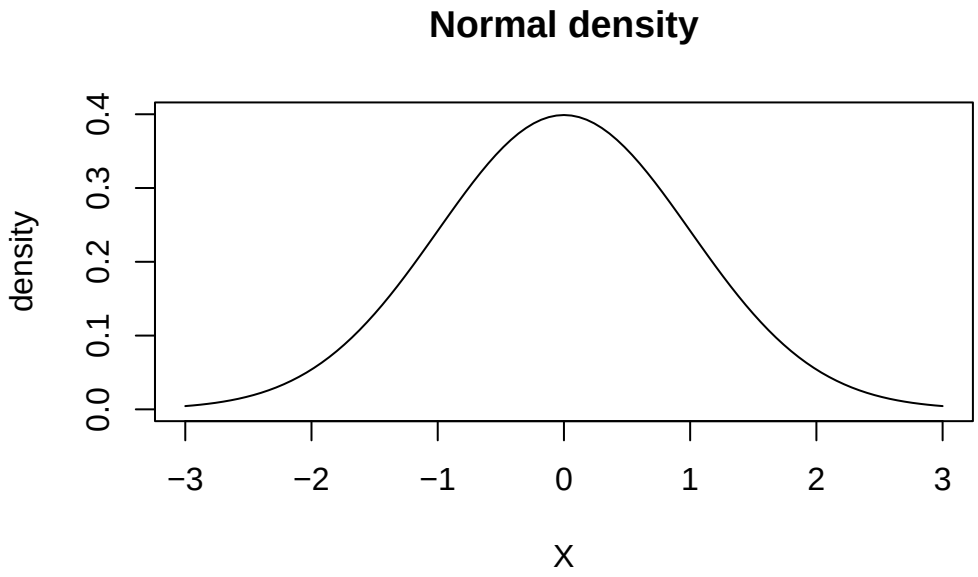
```r
rbinom(1, n=10, prob=0.5)
```

```
 [1] 1 0 0 1 0 0 0 1 1 0
```

Continuous data

- we cannot talk about the probability of each outcome
  - probability density function does not give you the probability of seeing an exact number, i.e. a point value (this is only possible in the discrete case)
  - in the continuous case: we talk about an area under the curve, there's an infinity of possible values between two values (which is different in the discrete case)
  - for example, P(x<1.69)
- the pdf of the normal distribution: $X\ norm(mean = \mu, sd = \sigma)$

```r
plot(function(x) dnorm(x, 0, 1), -3,3,
main = 'Normal density', ylim=c(0,.4),
ylab = 'density', xlab='X')
```



Probability of seeing possible range of values

```
pnorm(Inf) - pnorm(-Inf)
```

```
[1] 1
```

```
pnorm(2) - pnorm(-2)
```

```
[1] 0.9544997
```

```
pnorm(3) - pnorm(-3)
```

```
[1] 0.9973002
```

```
pnorm(0.5) - pnorm(-0.5)
```

```
[1] 0.3829249
```

Quantiles: what is the value (quantile x) of a $Normal(\mu, \sigma)$ such that $P(X < x) = 0.975$.

```r
qnorm(0.975, mean=500, sd=100)
```

```
[1] 695.9964
```

It is also possible to compute upper tail:

```r
pnorm(2, lower.tail=T)
```

```
[1] 0.9772499
```

```r
pnorm(2, lower.tail=F)
```

```
[1] 0.02275013
```

```r
1- pnorm(2, lower.tail=T)
```

```
[1] 0.02275013
```

Overview of functions:

- dnorm: the pdf
- pnorm: the cdf
- qnorm: the inverse of cdf
- for binomial: dbinom, pbinom, qbinom
- for t-distribution: dt, pt, qt

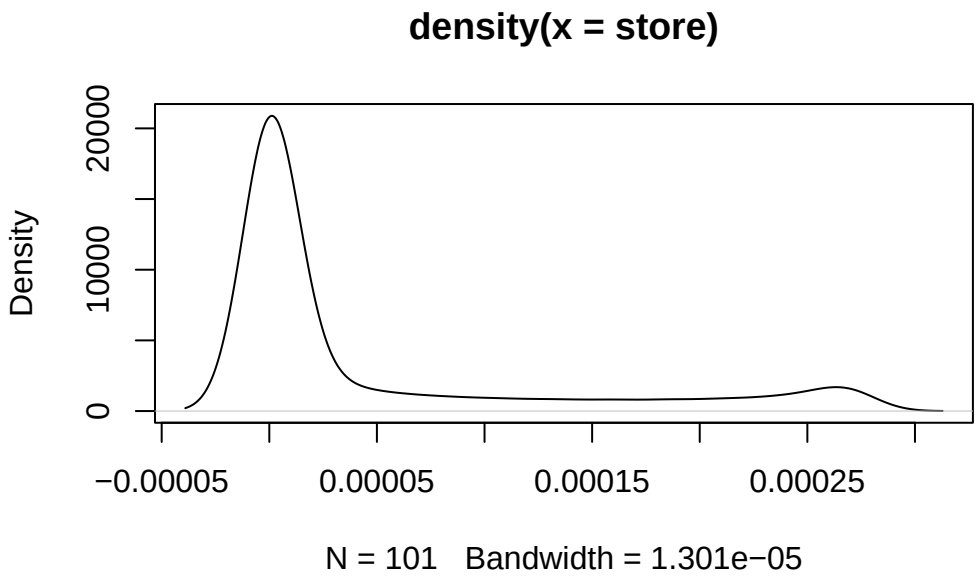
Maximum Likelihood Estimation

- sample mean (xbar), standard deviation (sd)
  - MLE of the true underlying distribution that we don't know
- joint probability of getting all these values
  - because they are independent: we can multiply all of the probabilities
  - the probability is a function of θ, this is the likelihood function
  - θ is the only variable that stays the same

Discrete case

```r
x <- c(5,4,3,5,2)
theta <- seq(0,1,by=0.01)
store <- rep(NA, length(theta))
for(i in 1:length(theta)){
    store[i] <- prod(dbinom(x, size=10, prob=theta[i]))
}

plot(density(store))
```



- maximum value on curve = MLE
- in the case of the binomial: $\theta = x$ divided by n
- in the *Continuous case*: sample mean and variance will give you MLE

Significance of the MLE

- having assumed a particular pdf, we can estimate the (unknown) parameters (mean and variance) of the distribution that generated our particular data -> distributional properties of the mean under repeated sampling

```r
abs(qt(0.025, df=15))
```

```
[1] 2.13145
```

Sampling distribution of the mean